STEADY LEVEL FLIGHT

Week 4: Steady level flight and trim

Find the elevator and tab settings that maintain steady level flight and remove the pilot's holding effort.

$$L = W, \qquad T = D, \qquad M = 0.$$

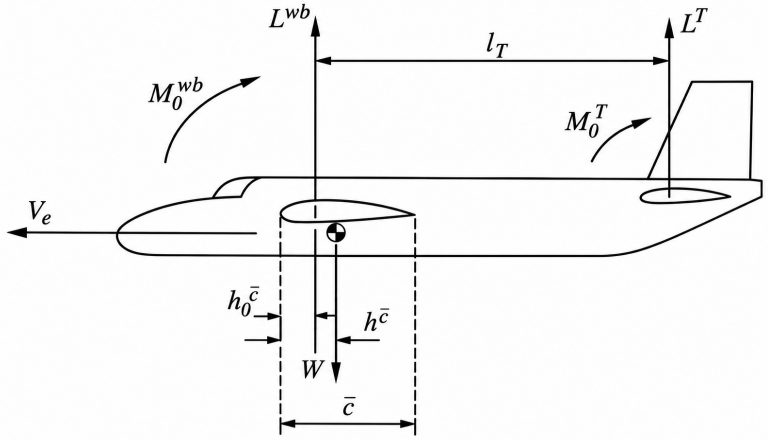


Figure notation: $L^{wb} = L_W$, $L^T = L_T$, $M_0^{wb} = M_0$, $V_e = V$. The separate tail couple M_0^T is neglected.

The pitching balance from Week 3

Take moments about the wing aerodynamic centre (AC), positive nose-up:

$$M = M_0 + W(h - h_0)\bar{c} - L_T l_T.$$

h and h_0 are the CG and wing-AC positions divided by the mean aerodynamic chord $\bar{c}$. The arm l_T runs from wing AC to tail AC.

$$C_M = \frac{M}{\frac{1}{2}\rho V^2 S_W \bar{c}}, \qquad C_L = C_W = \frac{W}{\frac{1}{2}\rho V^2 S_W}.$$

$$C_M = C_{M0} + (h - h_0)C_L - \bar{V}_T C_L^T = 0.$$

C_{M0} is the wing–body couple coefficient. C_L uses wing area S_W; C_L^T uses tail area S_T.

1 / PITCHING EQUILIBRIUM

Tail volume and the required tail lift

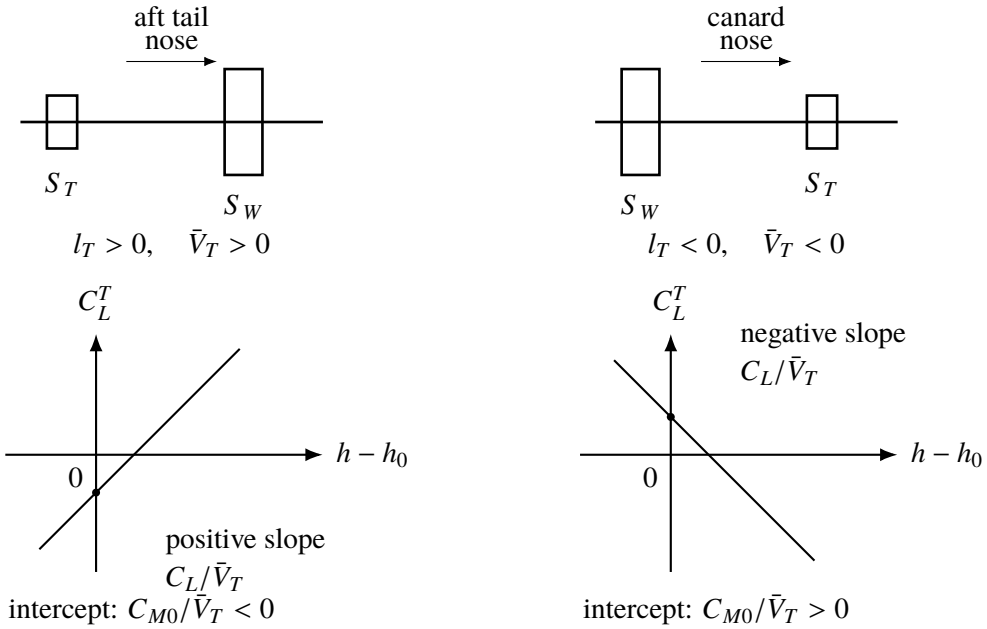
1. Rearrange the pitching balance

$$0 = C_{M0} + (h - h_0)C_L - \bar{V}_T C_L^T$$

$$C_L^T = \frac{h - h_0}{\bar{V}_T} C_L + \frac{C_{M0}}{\bar{V}_T}, \qquad \bar{V}_T = \frac{S_T}{S_W}\frac{l_T}{\bar{c}}.$$

Tail volume combines relative area and moment arm. The signed arm l_T is positive aft of the wing AC.

2. Compare aft-tail and canard configurations



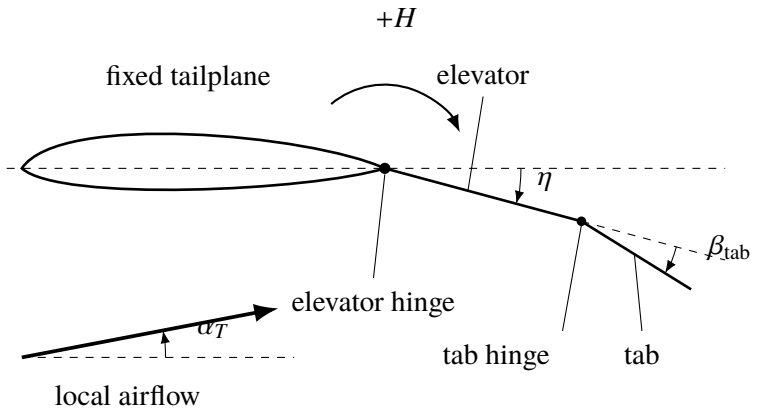
For fixed $C_L > 0$ and $C_{M0} < 0$:

$$\frac{\partial C_L^T}{\partial (h - h_0)} = \frac{C_L}{\bar{V}_T}, \qquad C_L^T\big|_{h=h_0} = \frac{C_{M0}}{\bar{V}_T}.$$

A longer arm or larger tail area needs a smaller magnitude of tail coefficient for the same balancing moment.

2 / THE ELEVATOR AND TAB

Tail lift and elevator hinge torque



Tail lift and hinge torque respond to the same three angles

$$C_L^T = a_1\alpha_T + a_2\eta + a_3\beta_{\text{tab}},$$

$$C_H = b_1\alpha_T + b_2\eta + b_3\beta_{\text{tab}}, \qquad C_H = \frac{H}{\frac{1}{2}\rho V^2 S_e \bar{c}_e}.$$

α_T	Tail incidence from its no-lift line to the local flow direction.
η	Elevator deflection relative to the fixed tailplane.
β_{tab}	Tab deflection relative to the elevator.

η and β_{tab} are positive trailing edge down; positive H tends to increase η.

a_1, a_2, a_3 are lift-response coefficients; b_1, b_2, b_3 are hinge-torque-response coefficients, here in $1/\text{rad}$.

S_e and $\bar{c}_e$ are the elevator reference area and chord, including the tab for the supplied hinge data.

2 / THE ELEVATOR AND TAB

Stick-fixed and stick-free trim

The aircraft must balance in both cases. The difference is the constraint on the elevator.

Stick-fixed	**Ideal stick-free**
The elevator angle is held.	The elevator can rotate.
The aircraft requires $C_M = 0$.	The aircraft requires $C_M = 0$.
A holding torque can balance the aerodynamic hinge torque.	No holding torque acts at the hinge.
The tab is held at a selected setting, often zero.	The tab is held at a selected setting; the elevator floats.

Balance the torques about the elevator hinge

$$H + \tau_{\text{hold}} = 0.$$

τ_{hold} is the torque transmitted through the control connection.

$$\begin{aligned} \text{Stick-fixed:} \quad & \tau_{\text{hold}} = -H, \\ \text{Ideal stick-free:} \quad & \tau_{\text{hold}} = 0 \quad \Longrightarrow \quad H = 0. \end{aligned}$$

Stick-fixed: $C_M = 0$.	Stick-free: $C_M = 0,\ C_H = 0$.

A held elevator can also have $C_H = 0$. In the ideal free case, friction, springs and other hinge torques are neglected.

3 / STICK-FIXED TRIM

Stick-fixed trim: derive the elevator angle

1. Express tail incidence in terms of aircraft lift

With $C_L = C_L^W + (S_T/S_W)C_L^T \simeq C_L^W = a_w\alpha$ and $\epsilon = (d\epsilon/d\alpha)\alpha$:

$$\begin{aligned}\alpha_T &= \alpha + \eta_T - \epsilon,\\ &= \frac{C_L}{a_w} + \eta_T - \frac{d\epsilon}{d\alpha}\frac{C_L}{a_w},\\ &= \frac{C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \eta_T.\end{aligned}$$

η_T is the tail setting relative to the wing no-lift line.

2. Substitute into the tail lift law

$$\begin{aligned}C_L^T &= a_1\alpha_T + a_2\eta + a_3\beta_{\text{tab}},\\ &= \frac{a_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + a_1\eta_T + a_2\eta + a_3\beta_{\text{tab}}.\end{aligned}$$

3. Substitute into the pitching moment equation

$$C_M = C_{M0} + (h - h_0)C_L - \bar{V}_T C_L^T.$$

$$\boxed{C_M = C_{M0} + (h - h_0)C_L - \bar{V}_T\left[\frac{a_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + a_1\eta_T + a_2\eta + a_3\beta_{\text{tab}}\right].}$$

4. Set $C_M = 0$ and isolate the elevator term

$$a_2\eta_{\text{trim}} = \frac{C_{M0}}{\bar{V}_T} + \frac{h - h_0}{\bar{V}_T}C_L - \frac{a_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) - a_1\eta_T - a_3\beta_{\text{tab}}.$$

Divide every term by a_2 to obtain η_{trim}.

3 / STICK-FIXED TRIM

Find the hinge torque at stick-fixed trim

1. Put the tail-incidence expression into the hinge law

$$\begin{aligned} C_H &= b_1\alpha_T + b_2\eta + b_3\beta_{\text{tab}}, \\ &= b_1\left[\frac{C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \eta_T\right] + b_2\eta + b_3\beta_{\text{tab}}, \\ &= \frac{b_1}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right)C_L + b_1\eta_T + b_2\eta + b_3\beta_{\text{tab}}. \end{aligned}$$

2. Write the trimmed elevator angle from page 5

$$\eta_{\text{trim}} = \frac{C_{M0}}{a_2\bar{V}_T} + \left[\frac{h - h_0}{a_2\bar{V}_T} - \frac{a_1}{a_2 a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right)\right]C_L - \frac{a_1}{a_2}\eta_T - \frac{a_3}{a_2}\beta_{\text{tab}}.$$

3. Substitute for η, then collect like terms

$$\begin{aligned} C_{H,\text{trim}} = {} & \frac{b_1}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right)C_L + b_1\eta_T + b_3\beta_{\text{tab}} \\ & + \frac{b_2 C_{M0}}{a_2\bar{V}_T} + \frac{b_2(h - h_0)}{a_2\bar{V}_T}C_L \\ & - \frac{b_2 a_1}{a_2 a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right)C_L - \frac{b_2 a_1}{a_2}\eta_T - \frac{b_2 a_3}{a_2}\beta_{\text{tab}}. \end{aligned}$$

$$\begin{aligned} C_{H,\text{trim}} = {} & \frac{b_2 C_{M0}}{a_2\bar{V}_T} \\ & + \left[\frac{b_2(h - h_0)}{a_2\bar{V}_T} + \frac{1}{a_w}\left(b_1 - \frac{b_2 a_1}{a_2}\right)\left(1 - \frac{d\epsilon}{d\alpha}\right)\right]C_L \\ & + \left(b_1 - \frac{b_2 a_1}{a_2}\right)\eta_T + \left(b_3 - \frac{b_2 a_3}{a_2}\right)\beta_{\text{tab}}. \end{aligned}$$

$$H_{\text{trim}} = \tfrac{1}{2}\rho V^2 S_e \bar{c}_e C_{H,\text{trim}}, \qquad \tau_{\text{hold}} = -H_{\text{trim}}.$$

4 / STICK-FREE TRIM

A free elevator changes the tail response

1. Let the elevator float: $C_H = 0$

$$b_1\alpha_T + b_2\eta + b_3\beta_{\text{tab}} = 0 \quad \Longrightarrow \quad \eta = -\frac{b_1}{b_2}\alpha_T - \frac{b_3}{b_2}\beta_{\text{tab}}.$$

Substitute $\alpha_T = (C_L/a_w)(1 - d\epsilon/d\alpha) + \eta_T$:

$$\eta_{\text{free}} = -\left[\frac{b_1 C_L}{b_2 a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \frac{b_1}{b_2}\eta_T + \frac{b_3}{b_2}\beta_{\text{tab}}\right].$$

2. Put this angle into the tail lift law

$$\begin{aligned} C_L^T &= a_1\alpha_T + a_2\eta + a_3\beta_{\text{tab}} \\ &= a_1\alpha_T + a_2\left(-\frac{b_1}{b_2}\alpha_T - \frac{b_3}{b_2}\beta_{\text{tab}}\right) + a_3\beta_{\text{tab}} \\ &= \left(a_1 - \frac{a_2 b_1}{b_2}\right)\alpha_T + \left(a_3 - \frac{a_2 b_3}{b_2}\right)\beta_{\text{tab}}. \end{aligned}$$

3. Name the combined response coefficients

$$\bar{a}_1 = a_1 - \frac{a_2 b_1}{b_2}, \qquad \bar{a}_3 = a_3 - \frac{a_2 b_3}{b_2}.$$

$$\boxed{C_L^T = \bar{a}_1\alpha_T + \bar{a}_3\beta_{\text{tab}}.}$$

The barred coefficients include the lift change caused by elevator motion. Their values depend on the supplied derivative signs.

4 / STICK-FREE TRIM

Stick-free trim: satisfy both moment balances

1. Use the free-elevator tail lift law

$$\begin{aligned} C_L^T &= \bar{a}_1 \alpha_T + \bar{a}_3 \beta_{\text{tab}}, \\ &= \frac{\bar{a}_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \bar{a}_1 \eta_T + \bar{a}_3 \beta_{\text{tab}}. \end{aligned}$$

2. Substitute into the aircraft pitching balance

$$C_{M,\text{free}} = C_{M0} + (h - h_0) C_L - \bar{V}_T C_L^T.$$

$$\boxed{C_{M,\text{free}} = C_{M0} + (h - h_0) C_L - \bar{V}_T \left[\frac{\bar{a}_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \bar{a}_1 \eta_T + \bar{a}_3 \beta_{\text{tab}}\right].}$$

Here $\bar{a}_1 = a_1 - a_2 b_1 / b_2$ and $\bar{a}_3 = a_3 - a_2 b_3 / b_2$.

3. Set $C_{M,\text{free}} = 0$ and isolate the tab term

$$\bar{a}_3 \beta_{\text{tab,trim}} = \frac{C_{M0}}{\bar{V}_T} + \frac{h - h_0}{\bar{V}_T} C_L - \frac{\bar{a}_1 C_L}{a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) - \bar{a}_1 \eta_T.$$

Divide by $\bar{a}_3$ for the tab angle; use $C_H = 0$ for the elevator angle:

$$\eta_{\text{free}} = -\left[\frac{b_1 C_L}{b_2 a_w}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \frac{b_1}{b_2}\eta_T + \frac{b_3}{b_2}\beta_{\text{tab,trim}}\right].$$

Alternatively, solve the two original laws together

$$\begin{aligned} a_2 \eta + a_3 \beta_{\text{tab}} &= C_L^T - a_1 \alpha_T, \\ b_2 \eta + b_3 \beta_{\text{tab}} &= -b_1 \alpha_T. \end{aligned}$$

The first supplies the required tail lift; the second imposes zero hinge torque.

5 / WORKED EXAMPLE

Question 1: find the tail load and incidence

Aircraft and aerodynamic data for Questions 1–3

$$\bar{V}_T = 0.8, \quad C_{M0} = -0.01, \quad h - h_0 = 0.1, \quad C_L = 1, \qquad \eta_T = 0, \quad \frac{d\epsilon}{d\alpha} = 0.4.$$

a_w	a_1	a_2	a_3	b_1	b_2	b_3
5.7	5	4	0.5	0.11	−0.12	−0.05

All seven derivatives are in 1/rad.

Question 1. Find the tail lift coefficient, downwash angle and tail incidence in steady level flight.

1. Use the aircraft pitching balance

$$C_L^T = \frac{C_{M0} + (h - h_0)C_L}{\bar{V}_T} = \frac{-0.01 + 0.1 \cdot 1}{0.8} \simeq 0.113.$$

2. Recover the wing incidence, then the downwash

$$\alpha \simeq \frac{C_L}{a_w} = \frac{1}{5.7} \simeq 0.175\,\text{rad} \quad (10.05^\circ),$$
$$\epsilon = \frac{d\epsilon}{d\alpha}\alpha = \frac{0.4}{5.7} \simeq 0.070\,\text{rad} \quad (4.02^\circ).$$

3. Find the incidence seen by the tail

$$\alpha_T = \alpha + \eta_T - \epsilon = \frac{1 - 0.4}{5.7} + 0 \simeq 0.105\,\text{rad} \quad (6.03^\circ).$$

The tail needs positive lift, but downwash reduces its incidence below the wing incidence.

5 / WORKED EXAMPLE

Question 2: trim with the tab at zero

Question 2. With the tab at zero, find the elevator angle and hinge-torque coefficient at trim.

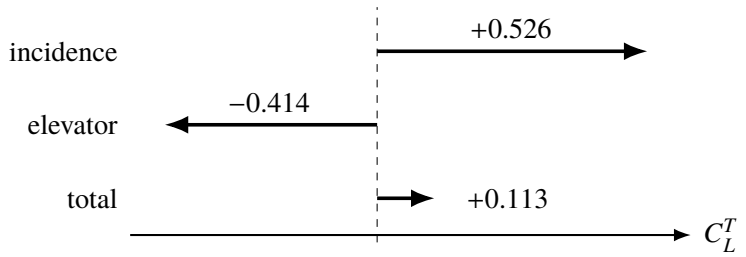
$$C_L^T = \frac{-0.01 + 0.1 \cdot 1}{0.8}, \qquad \alpha_T = \frac{0.6}{5.7}\,\text{rad}, \qquad \beta_{\text{tab}} = 0.$$

1. Set the elevator to produce the required tail coefficient

$$\begin{aligned} C_L^T &= a_1\alpha_T + a_2\eta + a_3\beta_{\text{tab}}, \\ \eta_{\text{trim}} &= \frac{C_L^T - a_1\alpha_T - a_3\beta_{\text{tab}}}{a_2} \\ &= \frac{(-0.01 + 0.1 \cdot 1)/0.8 - 5 \cdot (0.6/5.7)}{4} \\ &\simeq -0.103\,\text{rad} \quad (-5.93^\circ). \end{aligned}$$

2. Calculate the hinge torque at that same elevator angle

$$\begin{aligned} C_{H,\text{trim}} &= b_1\alpha_T + b_2\eta_{\text{trim}} + b_3\beta_{\text{tab}} \\ &= 0.11 \cdot \frac{0.6}{5.7} - 0.12 \cdot \eta_{\text{trim}} \simeq 2.40 \times 10^{-2}. \end{aligned}$$



The elevator moves up, reducing the positive tail lift to the required value. The hinge torque still needs an opposing holding torque.

5 / WORKED EXAMPLE

Question 3: trim with no holding torque

Question 3. Find the elevator and tab angles that preserve the required tail load with no holding torque.

1. Impose both the lift requirement and $C_H = 0$

$$\begin{aligned} a_2\eta + a_3\beta_{\text{tab}} &= C_L^T - a_1\alpha_T, \\ b_2\eta + b_3\beta_{\text{tab}} &= -b_1\alpha_T. \end{aligned} \qquad \alpha_T = \frac{0.6}{5.7}, \quad C_L^T = \frac{-0.01 + 0.1 \cdot 1}{0.8}.$$

$$\begin{aligned} 4\eta + 0.5\beta_{\text{tab}} &= C_L^T - 5\alpha_T, \\ -0.12\eta - 0.05\beta_{\text{tab}} &= -0.11\alpha_T. \end{aligned}$$

2. Multiply the second equation by 10, then add

$$-1.2\eta - 0.5\beta_{\text{tab}} = -1.1\alpha_T, \qquad 2.8\eta = C_L^T - 6.1\alpha_T.$$

$$\eta = \frac{C_L^T - 6.1\alpha_T}{2.8} \simeq -0.189\,\text{rad} \quad (-10.84^\circ).$$

3. Recover the tab angle from the hinge equation

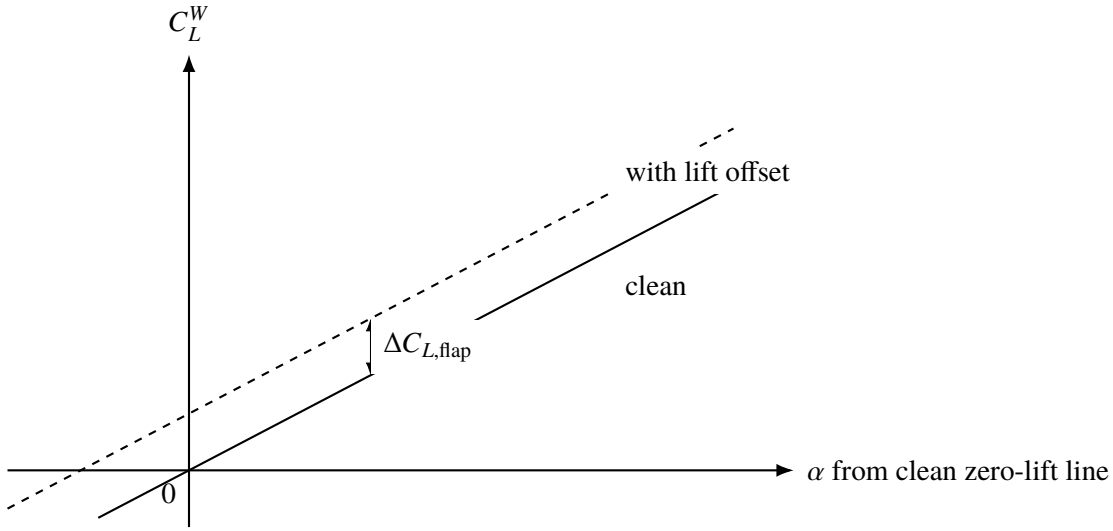
$$\beta_{\text{tab}} = \frac{0.11\alpha_T - 0.12\eta}{0.05} \simeq 0.686\,\text{rad} \quad (39.28^\circ).$$

Using unrounded angles gives $C_L^T \simeq 0.113$ and $C_H = 0$. The large tab angle is a formal linear-model result; it lies outside a credible small-deflection prediction.

6 / CHANGING THE FLIGHT CONDITION

Flaps change the incidence needed for lift

Keep α measured from the **clean-wing no-lift line**.



1. Include the additional wing lift at the same incidence

$$C_L^W = a_w \alpha + \Delta C_{L,\text{flap}}.$$

$\Delta C_{L,\text{flap}}$ is the flap-induced lift offset in this local linear model.

2. Find the incidence for the required lift

With $C_L \simeq C_L^W$,

$$\alpha \simeq \frac{C_L - \Delta C_{L,\text{flap}}}{a_w}.$$

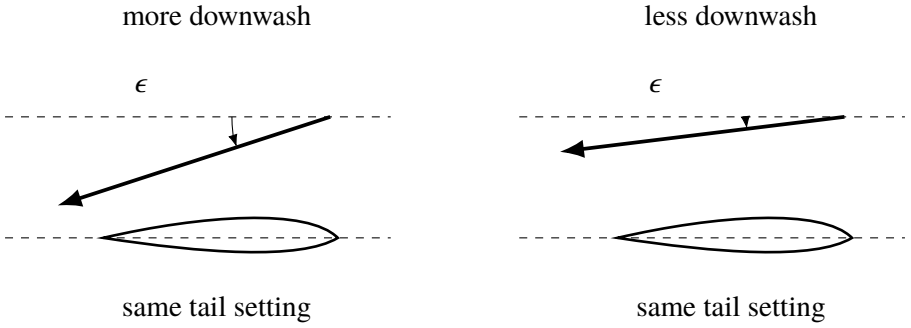
3. Use that incidence in the tail geometry

$$\alpha_T = \alpha + \eta_T - \epsilon.$$

Flaps can also change C_{M0} and downwash. Those changes require their own aerodynamic data before recalculating trim.

6 / CHANGING THE FLIGHT CONDITION

Changed downwash changes the trim setting



Consider a reduction in downwash, as can occur near the ground. Isolate this effect by holding α, η_T, β_{tab} and the required tail load fixed.

1. Find the change in tail incidence

$$\alpha_T = \alpha + \eta_T - \epsilon \quad \Longrightarrow \quad \Delta\alpha_T = -\Delta\epsilon.$$

2. Keep the tail lift coefficient unchanged

$$\Delta C_L^T = a_1 \Delta\alpha_T + a_2 \Delta\eta = 0, \qquad a_2 \Delta\eta = a_1 \Delta\epsilon.$$

$$\Delta\eta = \frac{a_1}{a_2} \Delta\epsilon.$$

Using $a_1 = 5\ 1/\text{rad}$, $a_2 = 4\ 1/\text{rad}$ and $\Delta\epsilon = -1°$:

$$\Delta\eta = \frac{5}{4} \cdot (-1°) = -1.25°.$$

Less downwash raises tail incidence. More up-elevator offsets that lift increase.

Conclusions

1. The aircraft moment balance sets the required tail load

$$C_M = C_{M0} + (h - h_0)C_L - \bar{V}_T C_L^T = 0, \qquad \bar{V}_T = \frac{S_T l_T}{S_W \bar{c}}.$$

Tail volume describes area and leverage, with the sign of the longitudinal arm.

2. Stick-fixed trim finds an elevator angle

$$\eta_{\text{trim}} = \frac{C_L^T - a_1\alpha_T - a_3\beta_{\text{tab}}}{a_2}, \qquad C_H = b_1\alpha_T + b_2\eta_{\text{trim}} + b_3\beta_{\text{tab}}.$$

The control connection supplies any required opposing hinge torque.

3. Stick-free trim also requires zero hinge torque

$$a_2\eta + a_3\beta_{\text{tab}} = C_L^T - a_1\alpha_T,$$
$$b_2\eta + b_3\beta_{\text{tab}} = -b_1\alpha_T.$$

The selected tab and floating elevator must satisfy both equations.

> Calculation order: required tail load ⟶ tail incidence ⟶ elevator and tab ⟶ check both balances.

The tutorial applies this sequence to different aircraft data and control settings.

Trim establishes equilibrium. Week 5 examines stability: the pitching response after a small disturbance.